REFORM ALLIANCE · FEDERAL SENTENCING DATA ANALYSIS

Who Receives Federal Probation?

What is linked to probation sentences in federal court, fiscal years 2014 to 2025

November 2026 · Probation Analysis Report · Companion to the Supervised Release Analysis Report

Probation is a sentence served in the community instead of prison. This report uses U.S. Sentencing Commission data to ask which people, offenses and places are linked to a higher or lower chance of receiving it. It compares people who are alike on the other characteristics in the data, and it checks whether the findings hold when missing information is handled in different ways.

5.7×
the odds of probation for fraud compared with drug trafficking, the largest offense difference

2.7×
the odds of probation for women compared with men

5.8%
of the leftover variation is tied to state and district, against 31% for supervised release

Key findings

1. **Offense type shows the largest differences.** Compared with drug trafficking, fraud had 5.7 times the odds of probation, other offenses 5.2 times, immigration 3.6 times and firearms 1.5 times. Sexual offenses had 84% lower odds and violent offenses 30% lower.
2. **Women and people with more education had higher odds.** Women had 2.7 times the odds of men, the opposite of supervised release. College graduates had 2.0 times the odds of people without a high school diploma.
3. **Non-citizens, Black and Hispanic defendants had lower odds.** Non-U.S. citizens had 77% lower odds, Black defendants 30% lower and Hispanic defendants 23% lower.
4. **Place matters much less than for supervised release.** About 5.8% of the leftover variation is tied to state and district, against 31% for supervised release. Case characteristics explain more of probation than place does.
5. **Probation became less common.** The odds did not change in FY2015 or FY2016, then fell to a low in FY2023 (0.72 vs. FY2014) and were 0.84 in FY2025. The results held up well when missing data were handled differently.

Read this first. These results describe patterns in recorded sentences. They do not show why a difference exists or whether it reflects unequal treatment. The models leave out criminal history, the guideline range, plea agreements and the facts of each case, because the dashboard data do not include them.

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1 Purpose and data

The analysis was designed to identify what is linked to higher use of federal probation, to inform efforts to reduce unnecessary use. It looks for differences between demographic groups, between states and between judicial districts.

- **Data.** Federal sentencing records from the U.S. Sentencing Commission, fiscal years 2014 to 2025, aggregated for the dashboard. Each row is a group of people who share the same characteristics and place, with counts of how many received each sentence type. In the probation data, 87% of rows describe a single person.
- **Who is counted.** Everyone sentenced, whether or not they were sentenced to prison.
- **Exclusions.** The Northern Mariana Islands were left out because there were too few cases.

Sample	
Rows for everyone sentenced	**554,676**
Rows used in the main analysis (no missing values on the variables in the model)	**514,157 (92.7%)**
Most often missing	**Education (7.1% of rows), race and ethnicity (1.6%), citizenship (0.9%), gender (0.3%)**
Never missing	**Offense category, fiscal year, judicial district**

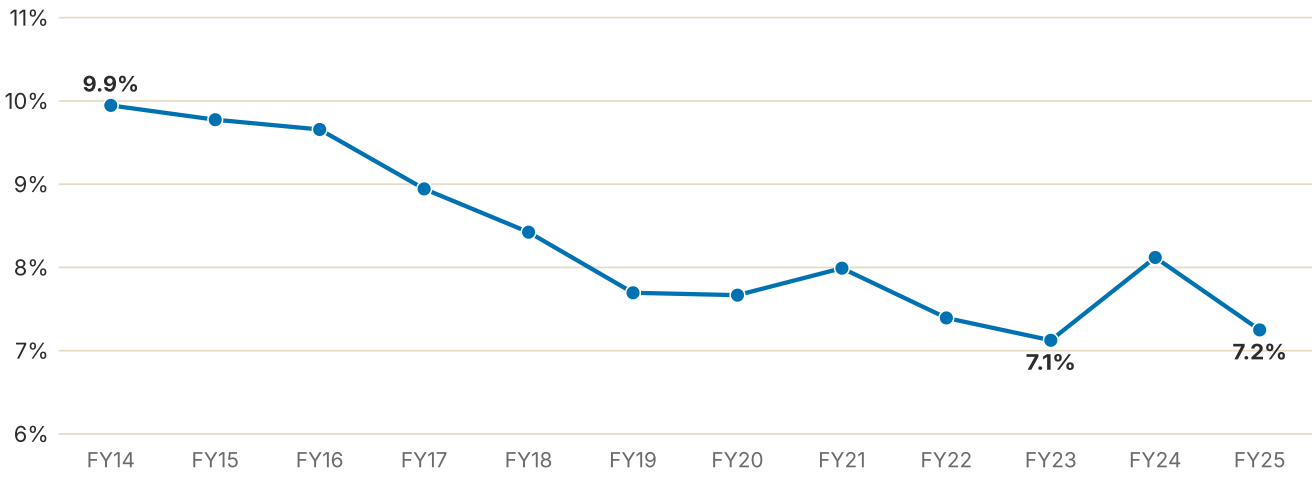


Figure 1. Probation rate over time. Share of everyone sentenced who received probation, national total, fiscal years 2014 to 2025. These are raw rates from the dashboard data. The rate was 9.9% in FY2014, fell to a low of 7.1% in FY2023 and was 7.2% in FY2025.

2 How to read the results

The outcome was modeled with a binomial mixed-effects (multilevel) logistic regression. In plain terms:

- For each row, the model uses the number of people who received probation and the number who did not.
- It estimates how the **odds** of probation differ between groups of people, holding the other characteristics in the model constant: age, race and ethnicity, gender, citizenship, education, offense type and fiscal year.
- It gives each state, and each district within a state, its own baseline level, so places can differ even when the people sentenced are alike.
- Results are odds ratios with 95% confidence intervals. An odds ratio of 1 means no difference. Below 1 means lower odds and above 1 means higher odds than the reference group.

Characteristic	Compared with (reference group)
Age	**Each additional year**
Race and ethnicity: Black, Hispanic, Other	**White**
Gender: Female	**Male**
Citizenship: Non-U.S. citizen	**U.S. citizen**
Education: H.S. graduate, Some college, College graduate	**Less than H.S. graduate**
Offense: Firearms, Fraud, Immigration, Sexual, Violent, Other	**Drug trafficking**
Fiscal year	**FY2014**

> **Odds are not percentages.** Odds compare the chance that something happens with the chance that it does not. Probation is uncommon: 8.4% of people sentenced received it. When an outcome is rare, odds and chances are close, so an odds ratio of 2.7 is close to 2.7 times the chance. In the raw data, 22% of women and 6% of men received probation. When an outcome is common, a large odds ratio can be a small change in percentage points.

No predictors overlapped strongly (all variance inflation factors were below 1.4). Because most rows describe a single person, the two-column binomial outcome handles the mix of single and grouped rows. Technical details are in the appendix.

3 What the model found

Who receives probation

Figure 2 (below) shows the adjusted odds ratio for every characteristic except age. The sections that follow describe each result.

Offense type

Offense type shows the largest differences. Compared with drug trafficking, fraud had 5.7 times the odds of probation (odds ratio 5.744), other offenses 5.2 times, immigration 3.6 times and firearms 1.5 times. Violent offenses had 30% lower odds and sexual offenses 84% lower, consistent with required prison terms for sexual offenses. For fraud, immigration and other offenses the pattern is the reverse of supervised release, where they had lower odds: offenses that are less likely to lead to supervised release after prison are more likely to lead to probation instead of prison.

Gender

Women had 2.7 times the odds of probation compared with men (odds ratio 2.692), the reverse of supervised release, where women had 13% lower odds. Women appear more likely to receive probation as an alternative to prison than supervised release after it.

Education

More education is linked to higher odds. Compared with people without a high school diploma, high school graduates had 37% higher odds, people with some college 85% and college graduates 99%. The step-up held under multiple imputation (odds ratio 1.91 for college graduates), unlike the education pattern for supervised release. One reading is that people with more education are seen as better candidates for a sentence in the community; the data cannot test that.

Citizenship, race and ethnicity

Non-U.S. citizens had 77% lower odds of probation than U.S. citizens who were otherwise similar (0.226), a smaller gap than for supervised release (95% lower). Compared with White defendants, Black defendants had 30% lower odds (0.701) and Hispanic defendants 23% lower (0.775); defendants in the Other group had 13% higher odds (1.128).

For supervised release, Black and White defendants showed no clear difference.

Age

Each additional year of age was linked to 0.5% higher odds of probation in the main model. The effect is small and disappears under multiple imputation (odds ratio 1.00).

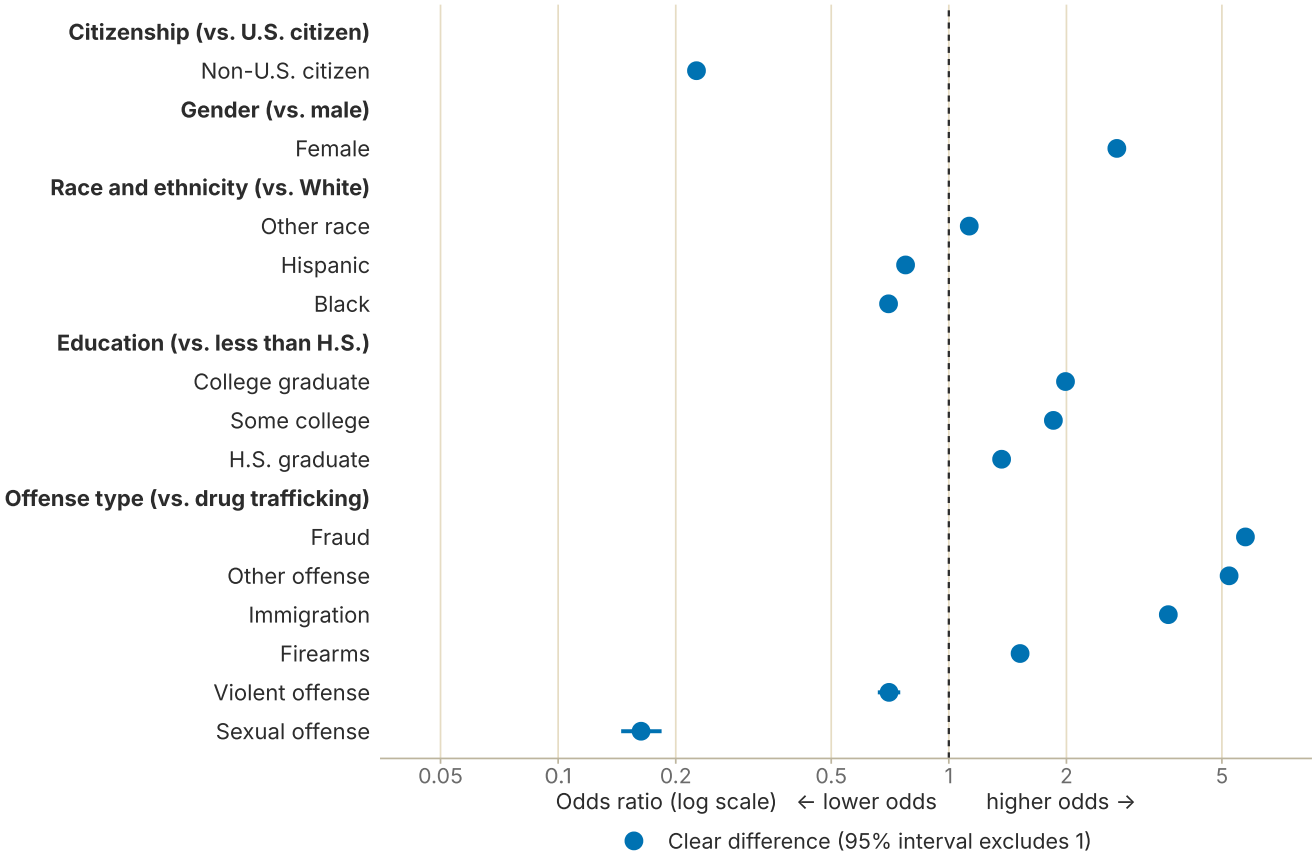


Figure 2. Adjusted odds ratios for probation. Odds ratios with 95% confidence intervals from the main model, adjusted for the other factors. Each point compares a group with the reference group named in its row heading. Every interval excludes 1, so every difference is clear. The horizontal axis is logarithmic.

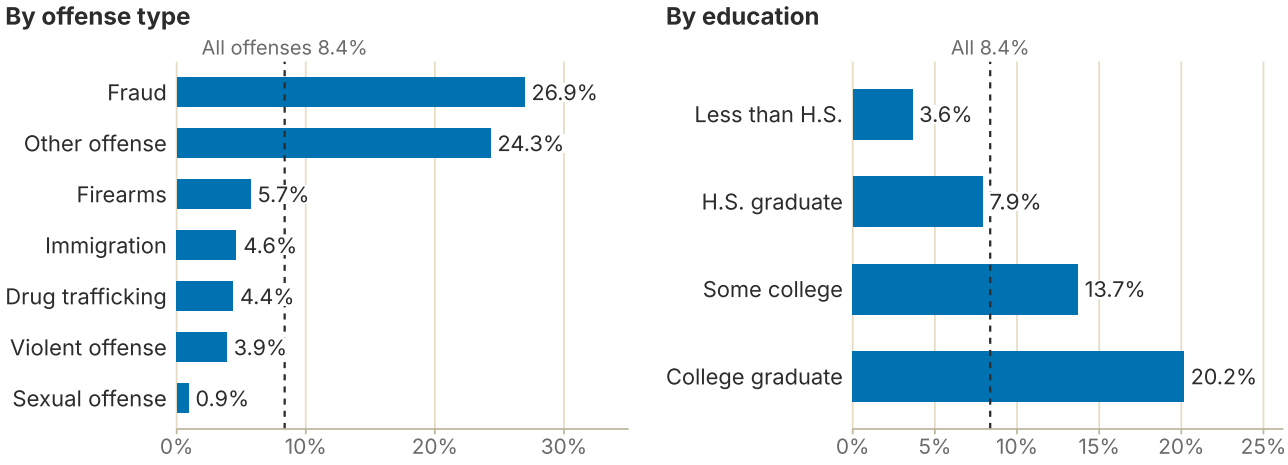


Figure 3. The raw rates behind the odds ratios. Unadjusted probation rates among everyone sentenced, national total, fiscal years 2014 to 2025. They do not hold other factors constant, which is why they can differ from the odds ratios. The dotted line is the rate across all people sentenced.

Change over time

The odds of probation fell over the period, while the odds of supervised release rose. In the main model the odds did not differ from FY2014 in FY2015 (1.00) or FY2016 (0.99). A decline began in FY2017 (0.93) and continued to a low in FY2023 (0.72), 28% lower than in FY2014. The odds were slightly higher in FY2024 (0.85) and lower again in FY2025 (0.84).

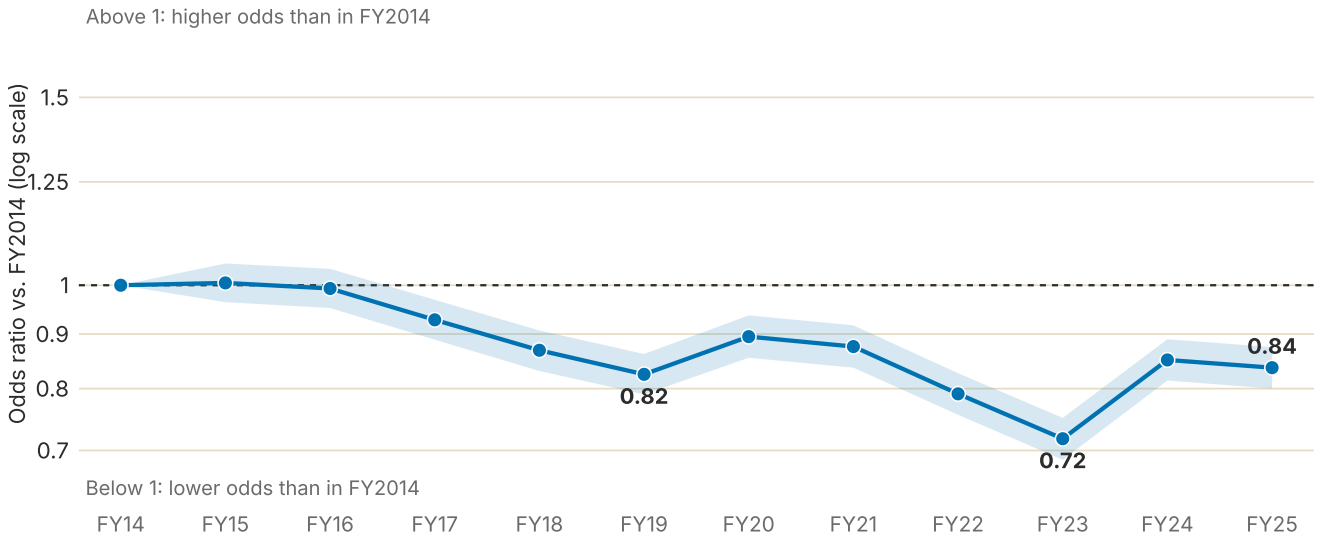


Figure 4. Odds of probation by fiscal year. Odds ratio for each fiscal year compared with FY2014, adjusted for the other factors in the model, with 95% confidence bands. The vertical axis is logarithmic.

Over most of this period federal sentencing shifted toward supervision after prison (supervised release) and away from community supervision without prison (probation). The decline held under both missing-data checks. The AR1 model, which lets each district follow its own year-to-year path, fit better than fiscal-year fixed effects (AIC lower by 1,619), by less than for supervised release, and left the results for individual characteristics essentially unchanged. National probation trends were very stable from year to year (AR1 correlation 0.897).

Where someone is sentenced

Place matters much less for probation than for supervised release. In the main model **5.8%** of the variation left after accounting for people's characteristics is tied to where they were sentenced: 0.8% to the state and 5.0% to the district within the state. For supervised release the combined figure is 31.1%. Districts vary more than states, but both vary little (district standard deviation 0.418, state 0.169).

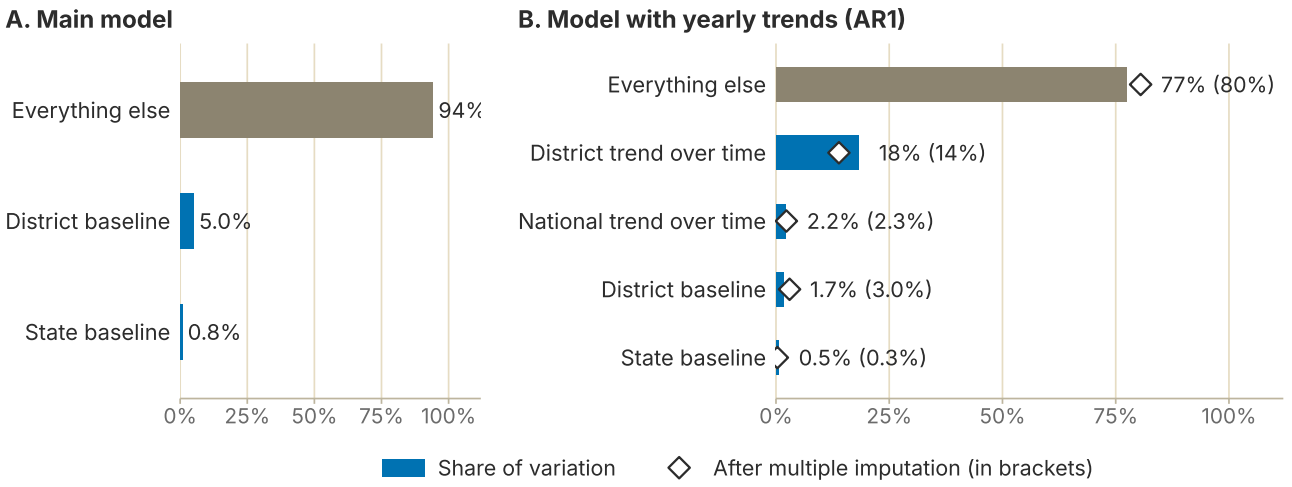


Figure 5. Where the variation in probation sits. Share of the total variation tied to each source, on the log-odds scale (intraclass correlation). Panel A is the main model. Panel B lets each district follow its own trend over time (AR1); diamonds show the same model after multiple imputation. The grey bar is everything else: differences between individual cases that place and year do not explain.

With district trends included, the district-level share rises to 20% (17% after imputation), still far below the 56% for supervised release. Most of the variation (77% in the yearly-trend model) is left to differences between individual cases. The analysis reads this as probation depending more on case characteristics than on place, so differences in probation may need to be addressed through changes that apply to specific kinds of cases rather than to particular districts. The model cannot see every case detail, so the remainder may include unmeasured features of the case.

4 How well the findings held up

The main model uses only rows with no missing values (92.7% of rows). Education and race are missing far more often for non-citizens and Hispanic defendants (Figure 6), so dropping incomplete rows risks selection bias. The model was therefore run twice more: once keeping people with missing values by treating missing as their own category, and once with multiple imputation (five filled-in datasets using random forests, combined with Rubin's rules).

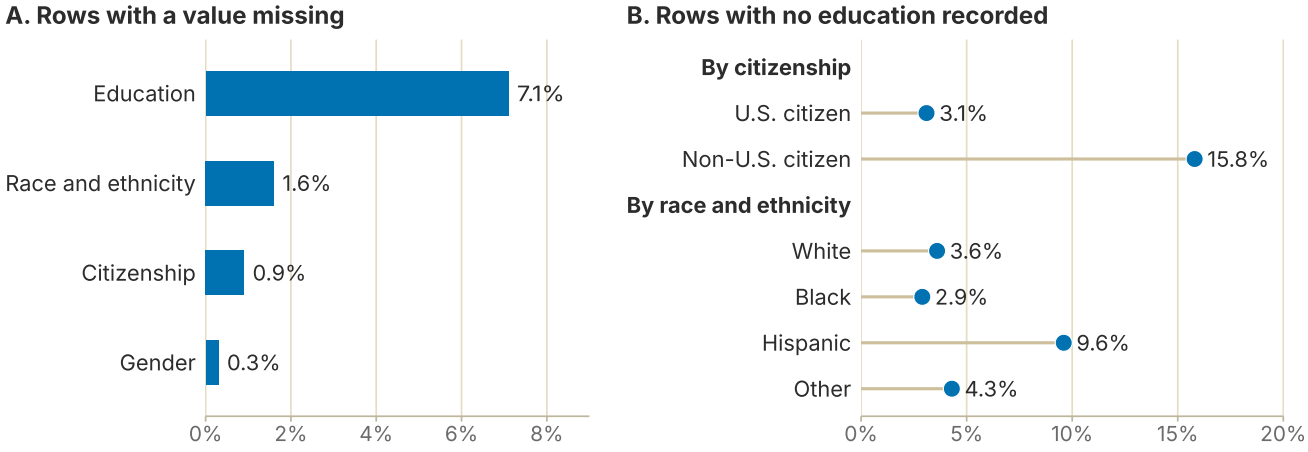


Figure 6. How much was missing, and where. Panel A: share of rows with a missing value for each field. Offense type, fiscal year and district had no missing values. Panel B: share of rows with no education recorded, by group.

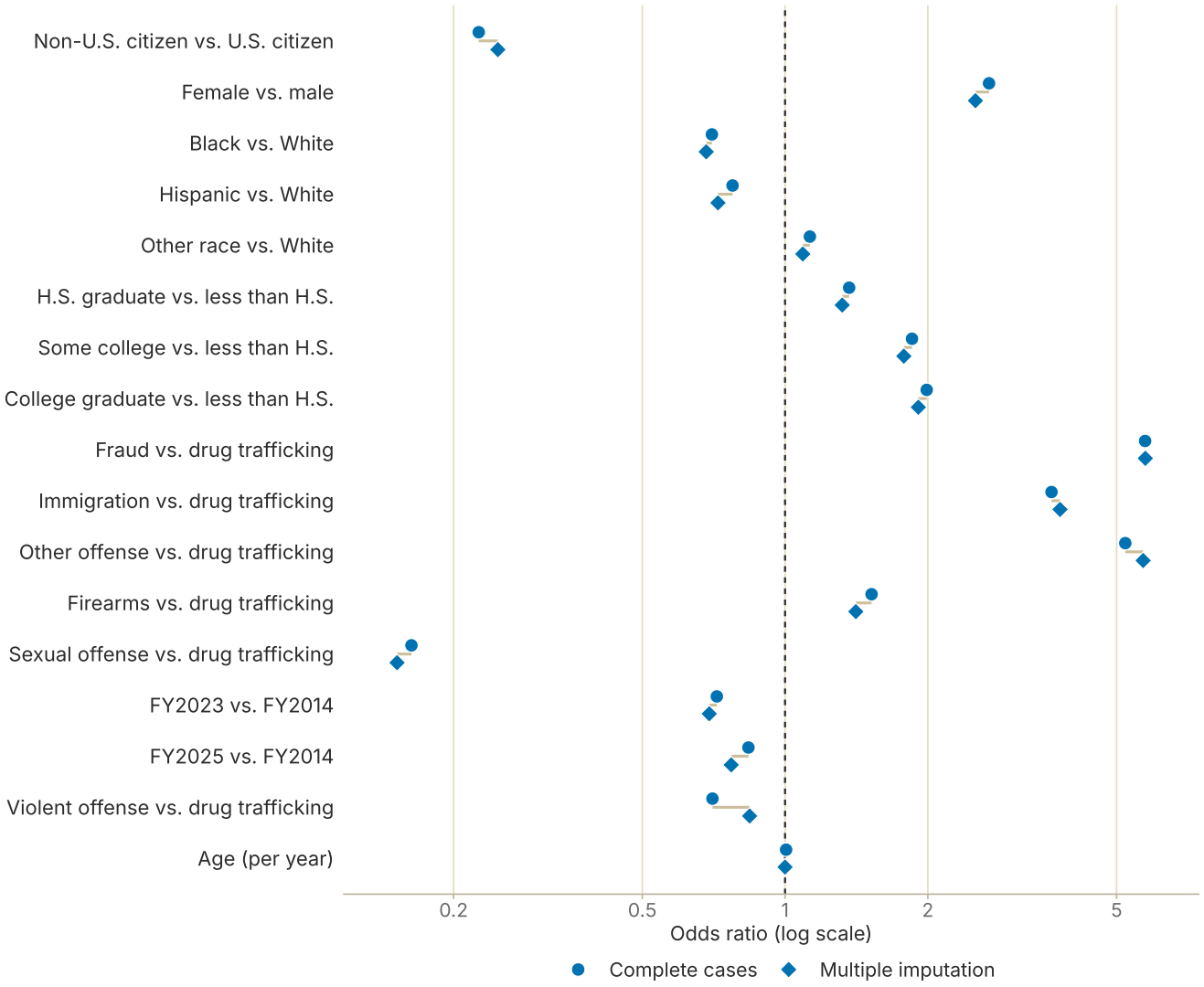


Figure 7. Odds ratios from the main model and after multiple imputation. Odds ratios for the same comparisons from the main model (complete cases) and after multiple imputation. Markers that stay close together show a result that holds. The horizontal axis is logarithmic.

- **Probation results were stable.** Under multiple imputation every result kept its direction, including the education pattern, which changed for supervised release. The violent offense gap shrank (odds ratio 0.703 to 0.842) and the small age effect disappeared.
- **Missing as its own category.** The citizenship gap grew (odds ratio 0.189). People with missing education had far higher odds of probation (3.965), the opposite of the supervised release data, where they had far lower odds, so the cases with missing education differ between the two datasets. People with missing race had 49% lower odds than White defendants. The decline in probation began slightly earlier in this check (FY2015 odds ratio 0.958).
- **Geography held:** the district-level share of variation was 17% after imputation, against 20% in the main yearly-trend model, and individual-level variation stayed dominant (80% against 77%).

5 Limits and open questions

What this analysis cannot tell you

- It describes patterns in recorded outcomes. It does not show why a difference exists, or whether it reflects unequal treatment.

- It leaves out things that influence sentences, such as criminal history, the guideline range, plea agreements and the facts of the case.
- Odds ratios are not differences in percentages. Probation is uncommon, so odds ratios and relative chances are close, but they are not the same.
- The estimates come from one set of data and one set of modeling choices. Confidence intervals are Wald intervals. Profile-likelihood intervals are recommended for final results.

Questions the analysis flags

- Which districts, if any, show the largest racial and ethnic gaps. One approach is to let the race effect vary by district.
- Whether the education pattern differs by offense type, for example through models run separately for each type.
- How the probation decline relates to the rise in supervised release over the same years.

How supervised release and probation compare

Comparison	Supervised release	Probation
Non-U.S. citizen vs. U.S. citizen	0.049 (95% lower odds)	0.226 (77% lower odds)
Female vs. male	0.867 (13% lower odds)	2.692 (2.7 times the odds)
Black vs. White	1.009 (no clear difference)	0.701 (30% lower odds)
Hispanic vs. White	0.828 (17% lower odds)	0.775 (23% lower odds)
College graduate vs. less than H.S.	0.897 (10% lower odds)	1.989 (99% higher odds)
Sexual offense vs. drug trafficking	4.413 (4.4 times the odds)	0.163 (84% lower odds)
Fraud vs. drug trafficking	0.386 (61% lower odds)	5.744 (5.7 times the odds)
Immigration vs. drug trafficking	0.153 (85% lower odds)	3.646 (3.6 times the odds)

Adjusted odds ratios from the two reports. The same groups often have opposite results: factors linked to more probation tend to be linked to less supervised release.

Source: U.S. Sentencing Commission individual offender datafiles for fiscal years 2014 to 2025, aggregated for the REFORM Alliance federal sentencing dashboard. Odds ratios are from binomial mixed-effects models fitted with the glmmTMB package in R.

A Appendix: tables and technical details

A1. Odds ratios from the main model

Group	Comparison	Odds ratio	95% CI	p-value	In plain words
Age	Age (per year)	1.005	1.005 to 1.006	<0.001	0.5% higher odds with each year of age
Citizenship	Non-U.S. citizen vs. U.S. citizen	0.226	0.218 to 0.235	<0.001	77% lower odds
Gender	Female vs. male	2.692	2.635 to 2.750	<0.001	2.7 times the odds
Race/ethnicity	Black vs. White	0.701	0.683 to 0.720	<0.001	30% lower odds
Race/ethnicity	Hispanic vs. White	0.775	0.751 to 0.799	<0.001	23% lower odds
Race/ethnicity	Other race vs. White	1.128	1.081 to 1.176	<0.001	13% higher odds
Education	H.S. graduate vs. less than H.S.	1.365	1.329 to 1.402	<0.001	37% higher odds
Education	Some college vs. less than H.S.	1.852	1.800 to 1.906	<0.001	85% higher odds
Education	College graduate vs. less than H.S.	1.989	1.917 to 2.063	<0.001	99% higher odds
Offense	Firearms vs. drug trafficking	1.522	1.467 to 1.579	<0.001	52% higher odds
Offense	Fraud vs. drug trafficking	5.744	5.577 to 5.917	<0.001	5.7 times the odds
Offense	Immigration vs. drug trafficking	3.646	3.500 to 3.799	<0.001	3.6 times the odds
Offense	Other offense vs. drug trafficking	5.217	5.060 to 5.380	<0.001	5.2 times the odds
Offense	Sexual offense vs. drug trafficking	0.163	0.145 to 0.184	<0.001	84% lower odds
Offense	Violent offense vs. drug trafficking	0.703	0.658 to 0.751	<0.001	30% lower odds

Binomial mixed-effects model, complete cases. Each odds ratio compares a group with its reference group, holding the other characteristics constant.

A2. Fiscal-year odds ratios

Fiscal year	Odds ratio	95% CI	p-value	In plain words
FY2015 vs. FY2014	1.005	0.964 to 1.048	0.798	No clear difference from FY2014
FY2016 vs. FY2014	0.993	0.952 to 1.036	0.746	No clear difference from FY2014
FY2017 vs. FY2014	0.928	0.889 to 0.969	<0.001	7.2% lower odds than in FY2014
FY2018 vs. FY2014	0.869	0.831 to 0.907	<0.001	13% lower odds than in FY2014
FY2019 vs. FY2014	0.825	0.790 to 0.862	<0.001	18% lower odds than in FY2014
FY2020 vs. FY2014	0.895	0.855 to 0.937	<0.001	11% lower odds than in FY2014
FY2021 vs. FY2014	0.876	0.837 to 0.917	<0.001	12% lower odds than in FY2014
FY2022 vs. FY2014	0.791	0.756 to 0.827	<0.001	21% lower odds than in FY2014
FY2023 vs. FY2014	0.718	0.686 to 0.751	<0.001	28% lower odds than in FY2014
FY2024 vs. FY2014	0.851	0.814 to 0.890	<0.001	15% lower odds than in FY2014
FY2025 vs. FY2014	0.837	0.800 to 0.875	<0.001	16% lower odds than in FY2014

A3. Where the variation sits

Source of variation	Main model	Model with yearly trends	Yearly-trend model, multiple imputation
State baseline	0.8%	0.5%	0.3%
District baseline (within state)	5.0%	1.7%	3.0%
State and district baselines combined	5.8%	2.3%	3.2%
District trend over time	–	18.2%	13.9%
National trend over time	–	2.2%	2.3%
District baseline plus district trend	–	19.9%	16.9%
Individual-level variation (remainder)	94.2%	77.4%	80.5%

Shares of total variation on the log-odds scale. The remainder row is one minus the other shares. Yearly-trend model components below.

Component	Std. dev. (complete cases)	Std. dev. (imputed)	AR1 correlation (complete cases)	AR1 correlation (imputed)
State baseline	0.151	0.102	–	–
District baseline	0.272	0.349	–	–
National trend	0.135	0.135	0.897	0.898
District trend	0.431	0.400	0.872	0.848

A4. Complete cases and multiple imputation

Comparison	Complete cases	Multiple imputation	Result
Non-U.S. citizen vs. U.S. citizen	0.226	0.248	Holds
Female vs. male	2.692	2.52	Holds
Black vs. White	0.701	0.682	Holds
Hispanic vs. White	0.775	0.722	Holds
Other race vs. White	1.128	1.09	Holds
H.S. graduate vs. less than H.S.	1.365	1.32	Holds
Some college vs. less than H.S.	1.852	1.78	Holds
College graduate vs. less than H.S.	1.989	1.91	Holds
Fraud vs. drug trafficking	5.744	5.75	Holds
Immigration vs. drug trafficking	3.646	3.80	Holds
Other offense vs. drug trafficking	5.217	5.69	Holds
Firearms vs. drug trafficking	1.522	1.41	Holds
Sexual offense vs. drug trafficking	0.163	0.152	Holds
FY2023 vs. FY2014	0.718	0.692	Holds
FY2025 vs. FY2014	0.837	0.770	Holds
Violent offense vs. drug trafficking	0.703	0.842	Same direction, smaller
Age (per year)	1.005	1.00	Effect disappears

Odds ratios.

A5. Model specification

- **Software.** glmmTMB in R. Outcome: `cbind(probation_sentence_flag, total_sentenced - probation_sentence_flag)`, binomial family with logit link. 86.8% of rows have `total_sentenced` equal to 1 (maximum 199, mean 1.43).
- **Fixed effects.** Age, race and ethnicity, gender, citizenship, education, offense category and fiscal year.
- **Random effects.** Nested random intercepts for judicial district within state, `(1 | state_name/state_district)`.
- **Yearly-trend (AR1) model.** Fiscal-year fixed effects replaced by a national AR1 term, `ar1(0 + year_f | dummy)`, and a district-level AR1 term, `ar1(0 + year_f | state_district)`, keeping the nested intercepts. AIC 311,921.9 against 313,541.3 for fixed-effect years. AR1 correlations: national 0.897, district 0.872 (after imputation 0.898 and 0.848). ICCs for AR1 terms use the variance $SD^2/(1-\rho^2)$.
- **Missing data.** Main model: complete cases. Check 1: missing treated as its own category (all 554,676 rows). Check 2: multiple imputation with miceRanger (random forest, m = 5, 50 trees, 5 iterations, seed 123) for race and ethnicity, gender, citizenship and education; the model was fitted to each imputed dataset and pooled with Rubin's rules.
- **Diagnostics.** All variance inflation factors below 1.4. Missingness was examined with the finalfit package.